Name : Mariam Mohamed Ramadan

Group :SE2

Lab 3

Part A: Linux Isolation

Task A1: Namespace Isolation Proof

```
root@96b573793060: /
debconf: (No usable dialog-like program is installed, so the dialog based frontend cannot be used. at /usr/share/perl5/
ebconf/FrontEnd/Dialog.pm line 78.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (Can't locate Term/ReadLine.pm in @INC (you may need to install the Term::ReadLine module) (@INC contains: /et
/perl /usr/local/lib/x86_64-linux-gnu/perl/5.34.0 /usr/local/share/perl/5.34.0 /usr/lib/x86_64-linux-gnu/perl5/5.34 /us
/share/perl5 /usr/lib/x86_64-linux-gnu/perl-base /usr/lib/x86_64-linux-gnu/perl/5.34 /usr/share/perl/5.34 /usr/local/li
/site_perl) at /usr/share/perl5/Debconf/FrontEnd/Readline.pm line 7.)
debconf: falling back to frontend: Teletype
Processing triggers for libc-bin (2.35-0ubuntu3.13) ...
root@96b573793060:/# hostname
96b573793060
root@96b573793060:/# ps -ef
UID        PID  PPID  C STIME TTY          TIME CMD
root         1     0  0 00:28 pts/0    00:00:00 bash
root       350     1  0 00:32 pts/0    00:00:00 ps -ef
root@96b573793060:/# ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0@if27: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state UP group default
    link/ether be:a3:b5:39:f9:bc brd ff:ff:ff:ff:ff:ff link-netnsid 0
    inet 172.17.0.2/16 brd 172.17.255.255 scope global eth0
       valid_lft forever preferred_lft forever
root@96b573793060:/# cat /proc/1/cgroup
0::/
root@96b573793060:/#
```

Task A2: Host vs. Container PID Mapping

```
mariamramadan@Ukhhh: ~
mariamramadan@Ukhhh:~$ docker ps
CONTAINER ID   IMAGE                     COMMAND           CREATED         STATUS                 PORTS
                         NAMES
96b573793060   ubuntu:22.04              "bash"            7 minutes ago   Up 7 minutes
                         nsdemo
f485fc15696e   week4-lab-order-service   "python app.py"   4 hours ago     Up 4 hours (healthy)   0.0.0.0:5002->5002/t
cp, [::]:5002->5002/tcp   order-service
76d1be50aeae   week4-lab-product-service "python app.py"   4 hours ago     Up 4 hours (healthy)   0.0.0.0:5001->5001/t
cp, [::]:5001->5001/tcp   product-service
mariamramadan@Ukhhh:~$ docker inspect nsdemo --format "{{.State.Pid}}"
53288
mariamramadan@Ukhhh:~$ ps -fp <THE_NUMBER_YOU_JUST_GOT>
-bash: syntax error near unexpected token `newline'
mariamramadan@Ukhhh:~$ ps -fp 53288
UID        PID  PPID  C STIME TTY          TIME CMD
mariamramadan@Ukhhh:~$
```

Task A3: Cgroup Resource Limiting

```
mariamramadan@Ukhhh: ~
CONTAINER ID   NAME            CPU %     MEM USAGE / LIMIT     MEM %     NET I/O           BLOCK I/O         PIDS
a333130c5233   tender_haibt    0.00%     18.66MiB / 256MiB     7.29%     50.8MB / 1.19MB   12.3kB / 121MB    1
f485fc15696e   order-service   0.02%     28.76MiB / 7.695GiB   0.37%     3.16kB / 2.3kB    12.3kB / 733kB    1
76d1be50aeae   product-service 0.02%     20.72MiB / 7.695GiB   0.26%     872B / 126B       12.3kB / 0B       1
```

Part B: Docker Image Layering

Task B1: Flask Application Setup

```
mariamramadan@Ukhhh:~$ mkdir lab3
mariamramadan@Ukhhh:~$ cd lab3
mariamramadan@Ukhhh:~/lab3$ echo "flask==3.0.0" > requirements.txt
mariamramadan@Ukhhh:~/lab3$ cat <<EOF > app.py
> from flask import Flask
Flask(> app = Flask(__name__)
@app.ro>
> @app.route("/")
> def home():
>     return "Hello from Lab 3"
app.rou>
> @app.route("/health")
> def health():
>     return "ok"
>
> if __name__ == "__main__":
app.run(>     app.run(host="0.0.0.0", port=5000)
OF> EOF
mariamramadan@Ukhhh:~/lab3$
```

Task B2: Basic Image Layering

```
> EOF
mariamramadan@Ukhhh:~/lab3$ docker build -f Dockerfile.basic -t lab3-basic .
[+] Building 28.2s (10/10) FINISHED                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                docker:default
=> [internal] load build definition from Dockerfile.basic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.2s
=> => transferring dockerfile: 1918                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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=> [internal] load metadata for docker.io/library/python:3.12-slim                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                1.6s
=> [internal] load .dockerignore                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.2s
=> => transferring context: 2B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> [1/5] FROM docker.io/library/python:3.12-slim@sha256:520153e2deb359602c9cffd84e491e3431d76e7bf95a3255c9ce9433b76ab99a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.3s
=> => resolve docker.io/library/python:3.12-slim@sha256:520153e2deb359602c9cffd84e491e3431d76e7bf95a3255c9ce9433b76ab99a                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.2s
=> [internal] load build context                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.3s
=> => transferring context: 318B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> CACHED [2/5] WORKDIR /app                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> [3/5] COPY requirements.txt .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> [4/5] RUN pip install -r requirements.txt                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                18.5s
=> [5/5] COPY app.py .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.3s
=> exporting to image                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => exporting layers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                3.5s
=> => exporting manifest sha256:e95e5b064836b49d7d92aaef22db17e73c32019303e65d36ce229186fb2296bd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> => exporting config sha256:ee7695491d2d40e1c238f30b9392722a468d24e5853df20a4f3b7870602d6092                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> => exporting attestation manifest sha256:37372203a13ccb9fbbbf87009dfd9185e7d52b1814a86eb0e4eb1c04570e05c47                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.1s
=> => exporting manifest list sha256:f94d8c7b64601c21ea56363cc7e3d4f65f0f6e3e8f5cdb0aee1e5981a8511abb                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => naming to docker.io/library/lab3-basic:latest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                0.0s
=> => unpacking to docker.io/library/lab3-basic:latest                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                2.0s
mariamramadan@Ukhhh:~/lab3$ docker history lab3-basic
IMAGE          CREATED        CREATED BY                                      SIZE      COMMENT
f94d8c7b6460   12 seconds ago   CMD ["python" "app.py"]                         0B        buildkit.dockerfile.v0
<missing>      12 seconds ago   EXPOSE [5000/tcp]                               0B        buildkit.dockerfile.v0
<missing>      12 seconds ago   COPY app.py . # buildkit                        12.3kB    buildkit.dockerfile.v0
<missing>      12 seconds ago   RUN /bin/sh -c pip install -r requirements.t…   16.5MB    buildkit.dockerfile.v0
<missing>      31 seconds ago   COPY requirements.txt . # buildkit              12.3kB    buildkit.dockerfile.v0
<missing>      4 hours ago      WORKDIR /app                                    8.19kB    buildkit.dockerfile.v0
<missing>      23 hours ago     CMD ["python3"]                                 0B        buildkit.dockerfile.v0
<missing>      23 hours ago     RUN /bin/sh -c set -eux;  for src in idle3 p…   16.4kB    buildkit.dockerfile.v0
<missing>      23 hours ago     RUN /bin/sh -c set -eux;   savedAptMark="$(a…   41.3MB    buildkit.dockerfile.v0
<missing>      23 hours ago     ENV PYTHON_SHA256=c08bc65a81971c1dd578318282…   0B        buildkit.dockerfile.v0
<missing>      23 hours ago     ENV PYTHON_VERSION=3.12.13                      0B        buildkit.dockerfile.v0
<missing>      23 hours ago     ENV GPG_KEY=7169605F62C751356D054A26A821E680…   0B        buildkit.dockerfile.v0
<missing>      23 hours ago     RUN /bin/sh -c set -eux;  apt-get update;  a…   4.94MB    buildkit.dockerfile.v0
<missing>      23 hours ago     ENV LANG=C.UTF-8                                0B        buildkit.dockerfile.v0
<missing>      23 hours ago     ENV PATH=/usr/local/bin:/usr/local/sbin:/usr…   0B        buildkit.dockerfile.v0
<missing>      2 days ago       # debian.sh --arch 'amd64' out/ 'trixie' '@1…   87.4MB    debuerreotype 0.17
mariamramadan@Ukhhh:~/lab3$
```

Activate Windows

Task B3: Multi-stage Optimization

```
mariamramadan@Ukhhh:~/lab3$ docker build -f Dockerfile.multistage -t lab3-multi .
[+] Building 24.7s (12/12) FINISHED                                                                                    docker:default
 => [internal] load build definition from Dockerfile.multistage                                                                   0.1s
 => => transferring dockerfile: 303B                                                                                              0.0s
 => [internal] load metadata for docker.io/library/python:3.12-slim                                                               2.6s
 => [internal] load .dockerignore                                                                                                 0.1s
 => => transferring context: 2B                                                                                                   0.0s
 => CACHED [builder 1/4] FROM docker.io/library/python:3.12-slim@sha256:520153e2deb359602c9cffd84e491e3431d76e7bf95a3255c9ce9433b76ab99a  0.2s
 => => resolve docker.io/library/python:3.12-slim@sha256:520153e2deb359602c9cffd84e491e3431d76e7bf95a3255c9ce9433b76ab99a         0.2s
 => [internal] load build context                                                                                                 0.1s
 => => transferring context: 63B                                                                                                  0.0s
 => CACHED [stage-1 2/4] WORKDIR /app                                                                                             0.0s
 => [builder 2/4] WORKDIR /build                                                                                                  0.3s
 => [builder 3/4] COPY requirements.txt .                                                                                         0.1s
 => [builder 4/4] RUN pip install --prefix=/install -r requirements.txt                                                          15.6s
 => [stage-1 3/4] COPY --from=builder /install /usr/local                                                                         0.5s
 => [stage-1 4/4] COPY app.py .                                                                                                   0.2s
 => exporting to image                                                                                                            1.9s
 => => exporting layers                                                                                                           1.0s
 => => exporting manifest sha256:8dba245f8943dfa958b060eed5415d4592276261c2c3734b324ed03b054da4e0                                 0.0s
 => => exporting config sha256:22e35935403463a746e3104704e0e44d0b443ce9ac9f0a8dd899eb5fe0f81d62a                                 0.0s
 => => exporting attestation manifest sha256:6bbc2315e5df02c72c9e6335fffdc70e5f1724bddcd2c7d548ac926b9a225dcf                     0.1s
 => => exporting manifest list sha256:0b91d71d2effbaa8f293c59b3ff3be15f472021f9a43eca788a1cd3cd78820c6                            0.0s
 => => naming to docker.io/library/lab3-multi:latest                                                                              0.0s
 => => unpacking to docker.io/library/lab3-multi:latest                                                                           0.6s
mariamramadan@Ukhhh:~/lab3$ docker image ls
                                                                                                          i Info →   U  In Use
IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
lab3-basic:latest                    f94d8c7b6460       199MB           49MB
lab3-multi:latest                    0b91d71d2eff       184MB         44.7MB
quay.io/coreos/etcd:v3.5.14          42106a001d1b      83.9MB         21.3MB
redis:7                              3e0669e42d4f       175MB         47.5MB
ubuntu:22.04                         962f6cadeae0       119MB         31.7MB
ubuntu:latest                        c4a8d5503dfb       119MB         31.7MB    U
week4-lab-order-service:latest       af94ab1bc7f7       202MB         49.2MB    U
week4-lab-product-service:latest     6216b63b864e       197MB         48.1MB    U
mariamramadan@Ukhhh:~/lab3$
```

Part C: Kubernetes Orchestration

Tasks C1 & C2 : Cluster Provisioning and Image Sideloading

```
mariamramadan@Ukhhh:~/lab3$ kind create cluster --name lab3-cluster
Creating cluster "lab3-cluster" ...
 ✓ Ensuring node image (kindest/node:v1.27.3) 🖼
 ✓ Preparing nodes 📦
 ✓ Writing configuration 📜
 ✓ Starting control-plane 🕹️
 ✓ Installing CNI 🔌
 ✓ Installing StorageClass 💾
Set kubectl context to "kind-lab3-cluster"
You can now use your cluster with:

kubectl cluster-info --context kind-lab3-cluster

Have a question, bug, or feature request? Let us know! https://kind.sigs.k8s.io/#community 🙂
mariamramadan@Ukhhh:~/lab3$ kind load docker-image lab3-multi --name lab3-cluster
Image: "lab3-multi" with ID "sha256:0b91d71d2effbaa8f293c59b3ff3be15f472021f9a43eca788a1cd3cd78820c6" not yet present on node "lab3-cluster-control-plane", loading...
mariamramadan@Ukhhh:~/lab3$ cat <<EOF > deployment.yaml
> apiVersion: apps/v1
> kind: Deployment
ata:
  > metadata:
>   name: lab3-web
 replica> spec:
>   replicas: 3
>   selector:
atchLab>     matchLabels:
>       app: lab3-web
>   template:
>     metadata:
>       labels:
>         app: lab3-web
      co>     spec:
ntainers>       containers:
>       - name: web
      im>         image: lab3-multi
>         imagePullPolicy: Never
    port>         ports:
>         - containerPort: 5000
> EOF
mariamramadan@Ukhhh:~/lab3$ cat <<EOF > service.yaml
> apiVersion: v1
> kind: Service
> metadata:
>   name: lab3-web-svc
  select> spec:
>   selector:
```

Activate Windows
Go to Settings to activate Windows.

Task C3: Replicated Deployment & Service

```
mariamramadan@Ukhhh: ~/lab3
mariamramadan@Ukhhh:~/lab3$ cat <<EOF > deployment.yaml
> apiVersion: apps/v1
> kind: Deployment
ata:
  > metadata:
>   name: lab3-web
 replica> spec:
>   replicas: 3
>   selector:
atchLab>     matchLabels:
>       app: lab3-web
>   template:
>     metadata:
>       labels:
>         app: lab3-web
      co>     spec:
ntainers>       containers:
>       - name: web
      im>         image: lab3-multi
>         imagePullPolicy: Never
    port>         ports:
>         - containerPort: 5000
> EOF
mariamramadan@Ukhhh:~/lab3$ cat <<EOF > service.yaml
> apiVersion: v1
> kind: Service
> metadata:
>   name: lab3-web-svc
  select> spec:
>   selector:
>     app: lab3-web
>   ports:
>     - port: 80
>       targetPort: 5000
e: Clu>   type: ClusterIP
> EOF
mariamramadan@Ukhhh:~/lab3$ kubectl apply -f deployment.yaml
deployment.apps/lab3-web created
mariamramadan@Ukhhh:~/lab3$ kubectl apply -f service.yaml
service/lab3-web-svc created
mariamramadan@Ukhhh:~/lab3$ kubectl get pods -o wide
NAME                        READY   STATUS    RESTARTS   AGE   IP           NODE                        NOMINATED NODE   READINESS GATES
lab3-web-58cdcb9c45-44dg8   1/1     Running   0          24s   10.244.0.7   lab3-cluster-control-plane   <none>           <none>
lab3-web-58cdcb9c45-64cgs   1/1     Running   0          24s   10.244.0.6   lab3-cluster-control-plane   <none>           <none>
lab3-web-58cdcb9c45-rqwqj   1/1     Running   0          24s   10.244.0.5   lab3-cluster-control-plane   <none>           <none>
mariamramadan@Ukhhh:~/lab3$
```

Task C4: Service Port Forwarding

```
ntainers>       containers:
>       - name: web
      im>         image: lab3-multi
>         imagePullPolicy: Never
    port>         ports:
>         - containerPort: 5000
> EOF
mariamramadan@Ukhhh:~/lab3$ cat <<EOF > service.yaml
> apiVersion: v1
> kind: Service
> metadata:
>   name: lab3-web-svc
  select> spec:
>   selector:
>     app: lab3-web
>   ports:
>     - port: 80
>       targetPort: 5000
e: Clu>   type: ClusterIP
> EOF
mariamramadan@Ukhhh:~/lab3$ kubectl apply -f deployment.yaml
deployment.apps/lab3-web created
mariamramadan@Ukhhh:~/lab3$ kubectl apply -f service.yaml
service/lab3-web-svc created
mariamramadan@Ukhhh:~/lab3$ kubectl get pods -o wide
NAME                        READY   STATUS    RESTARTS   AGE   IP           NODE                        NOMINATED NODE   READINESS GATES
lab3-web-58cdcb9c45-44dg8   1/1     Running   0          24s   10.244.0.7   lab3-cluster-control-plane   <none>           <none>
lab3-web-58cdcb9c45-64cgs   1/1     Running   0          24s   10.244.0.6   lab3-cluster-control-plane   <none>           <none>
lab3-web-58cdcb9c45-rqwqj   1/1     Running   0          24s   10.244.0.5   lab3-cluster-control-plane   <none>           <none>
mariamramadan@Ukhhh:~/lab3$ kubectl get pods -o wide
NAME                        READY   STATUS    RESTARTS   AGE     IP           NODE                        NOMINATED NODE   READINESS GATES
lab3-web-58cdcb9c45-44dg8   1/1     Running   0          5m59s   10.244.0.7   lab3-cluster-control-plane   <none>           <none>
lab3-web-58cdcb9c45-64cgs   1/1     Running   0          5m59s   10.244.0.6   lab3-cluster-control-plane   <none>           <none>
lab3-web-58cdcb9c45-rqwqj   1/1     Running   0          5m59s   10.244.0.5   lab3-cluster-control-plane   <none>           <none>
mariamramadan@Ukhhh:~/lab3$ kubectl port-forward service/lab3-web-svc 8080:80
Forwarding from 127.0.0.1:8080 -> 5000
Forwarding from [::1]:8080 -> 5000
curl http://localhost:8080/
```

Part D: Scheduling

Task D1 & D2: Node Metadata Labeling & Targeted Pod Placement

```
mariamramadan@Ukhhh: ~
>       labels:
>         app: lab3-web
>     spec:
>       nodeSelector:
>         node-role: general
>       containers:
>       - name: web
>         image: lab3-multi
>         imagePullPolicy: Never
>         ports:
>         - containerPort: 5000
> EOF
mariamramadan@Ukhhh:~$ kubectl apply -f deployment.yaml
deployment.apps/lab3-web configured
mariamramadan@Ukhhh:~$ kubectl get pods -o wide
NAME                          READY   STATUS              RESTARTS   AGE   IP           NODE                       NOMIN
ATED NODE   READINESS GATES
lab3-web-58cdcb9c45-44dg8     1/1     Running             0          14m   10.244.0.7   lab3-cluster-control-plane   <none
>           <none>
lab3-web-58cdcb9c45-64cgs     1/1     Terminating         0          14m   10.244.0.6   lab3-cluster-control-plane   <none
>           <none>
lab3-web-58cdcb9c45-rqwqj     1/1     Terminating         0          14m   10.244.0.5   lab3-cluster-control-plane   <none
>           <none>
lab3-web-8457f846b7-2s77f     1/1     Running             0          8s    10.244.0.8   lab3-cluster-control-plane   <none
>           <none>
lab3-web-8457f846b7-67zhx     1/1     Running             0          4s    10.244.0.9   lab3-cluster-control-plane   <none
>           <none>
lab3-web-8457f846b7-6rts4     0/1     ContainerCreating   0          1s    <none>       lab3-cluster-control-plane   <none
>           <none>
mariamramadan@Ukhhh:~$
```

Part E: Self-Healing

Task E1: Automated Pod Recovery

```
>           <none>
mariamramadan@Ukhhh:~$ kubectl delete pod
error: resource(s) were provided, but no name was specified
mariamramadan@Ukhhh:~$ kubectl delete pod mariam
Error from server (NotFound): pods "mariam" not found
mariamramadan@Ukhhh:~$ kubectl get pods
NAME                          READY   STATUS    RESTARTS   AGE
lab3-web-8457f846b7-2s77f     1/1     Running   0          3m14s
lab3-web-8457f846b7-67zhx     1/1     Running   0          3m10s
lab3-web-8457f846b7-6rts4     1/1     Running   0          3m7s
mariamramadan@Ukhhh:~$ kubectl delete pod lab3-web-8457f846b7-2s77f
pod "lab3-web-8457f846b7-2s77f" deleted from default namespace
kubectl get pods -w
mariamramadan@Ukhhh:~$ kubectl get pods -w
NAME                          READY   STATUS    RESTARTS   AGE
lab3-web-8457f846b7-67zhx     1/1     Running   0          4m7s
lab3-web-8457f846b7-6rts4     1/1     Running   0          4m4s
lab3-web-8457f846b7-7thw2     1/1     Running   0          34s
```

Task E2 & E3 : Health Probe Implementation & Process-Level Self-Healing

```
> EOF
mariamramadan@Ukhhh:~$ kubectl apply -f probe-deployment.yaml
deployment.apps/lab3-web-probe created
mariamramadan@Ukhhh:~$ kubectl get pods
NAME                              READY   STATUS    RESTARTS   AGE
lab3-web-8457f846b7-67zhx         1/1     Running   0          9m5s
lab3-web-8457f846b7-6rts4         1/1     Running   0          9m2s
lab3-web-8457f846b7-7thw2         1/1     Running   0          5m32s
lab3-web-probe-6788c59d-vkkhm     0/1     Running   0          10s
lab3-web-probe-6788c59d-vq5fg     0/1     Running   0          10s
mariamramadan@Ukhhh:~$ kubectl exec -it lab3-web-8457f846b7-7thw2 -- /bin/sh
# kill 1
# kubectl get pods
/bin/sh: 2: kubectl: not found
```

```
# exit
command terminated with exit code 127
mariamramadan@Ukhhh:~$ kubectl get pods
NAME                              READY   STATUS    RESTARTS   AGE
lab3-web-8457f846b7-67zhx         1/1     Running   0          12m
lab3-web-8457f846b7-6rts4         1/1     Running   0          12m
lab3-web-8457f846b7-7thw2         1/1     Running   0          8m33s
lab3-web-probe-6788c59d-vkkhm     1/1     Running   0          3m11s
lab3-web-probe-6788c59d-vq5fg     1/1     Running   0          3m11s
mariamramadan@Ukhhh:~$
```

D5: Concept Reflections :

1. Namespaces vs. Resource Use: Namespaces only isolate visibility (what a process sees). Without cgroups, a container could still consume 100% of the host's CPU/RAM and starve others.

2. Cgroups & Stability: They enforce resource limits. This prevents a "noisy neighbor" or a buggy app from exhausting host resources and crashing the entire node.

3. Image Layering in CI/CD: It enables caching. Docker only rebuilds changed layers, making builds faster and reducing storage/bandwidth for deployments.

4. Desired State: The configuration defined in your YAML (e.g., 3 replicas). Kubernetes constantly adjusts the Actual State to match this target.

5. Self-healing & Overhead: It automates recovery. Kubernetes detects failures via probes and replaces dead pods instantly without requiring manual human intervention.

6. Readiness vs. Liveness: Readiness tells the Service if the pod can accept traffic; Liveness tells the cluster if the pod is "alive" or needs to be restarted.

7. Multi-node Scheduling: In a multi-node cluster, the scheduler distributes pods across different physical servers for better load balancing and high availability

13.Clean Up :

```
mariamramadan@Ukhhh:~$ kind delete cluster --name lab3-cluster
Deleting cluster "lab3-cluster" ...
Deleted nodes: ["lab3-cluster-control-plane"]
mariamramadan@Ukhhh:~$ docker rmi lab3-basic lab3-multi
Untagged: lab3-basic:latest
Deleted: sha256:f94d8c7b64601c21ea56363cc7e3d4f65f0f6e3e8f5cdb0aee1e5981a8511abb
Untagged: lab3-multi:latest
Deleted: sha256:0b91d71d2effbaa8f293c59b3ff3be15f472021f9a43eca788a1cd3cd78820c6
mariamramadan@Ukhhh:~$
```